

Expt. No. : 3

Date :

BUILD A FUNCTION IN PYTHON TO CHECK DUPLICATE LETTERS

AIM

To Write a Python program to check whether any word in a given sentence contains duplicate characters or not.

SOFTWARE REQUIRED

Python 3.14

THEORY

In text processing and data validation, it is often necessary to analyse the structure and content of strings. One common requirement is to check for duplicate characters within words of a sentence. This can be important for various applications, including data cleaning, linguistic analysis, and coding challenges

PROBLEM

Build a function in Python to compare two words and determine whether they share any common letters. The function should return True if at least one common letter exists; otherwise, return False.

APPLICATION PROBLEM

During electrical panel documentation, panel names are entered as a sentence for verification. Build a Python function to detect duplicate letters in any panel name and return True if found; otherwise, return False.

ALGORITHM

STEP 1: Start

STEP 2: Input: String (the input string containing characters)

STEP 3: Initialize an empty dictionary to store character frequencies: `char_freq = {}`

STEP 4: Loop through each character in the string:

a. If the character is already in `char_freq`: i. Increment the frequency count for that character

b. Else:

STEP 5: i Add the character to `char_freq` with a frequency count of 1

STEP 6: Initialize an empty list to store duplicate characters: `duplicates = []`

STEP 7: Loop through each key-value pair in `char_freq`:

a. If the frequency count is greater than 1: i. Append the character to the `duplicates` list

STEP 8: Construct a statement using the `duplicates` list:

a. If the `duplicates` list is not empty: i. Construct a statement listing the duplicate characters

b. Else: i. Output a message indicating that no duplicate characters were found

STEP 9: Output the constructed statement

STEP 10: End

PROGRAM CODING

```

def common_letters(word1, word2):
    for i in word1:
        if i in word2:
            return True
    return False
word1 = input("Enter first word: ")
word2 = input("Enter second word: ")
print(common_letters(word1, word2))

```

OUTCOMES OF PROGRAM CODING

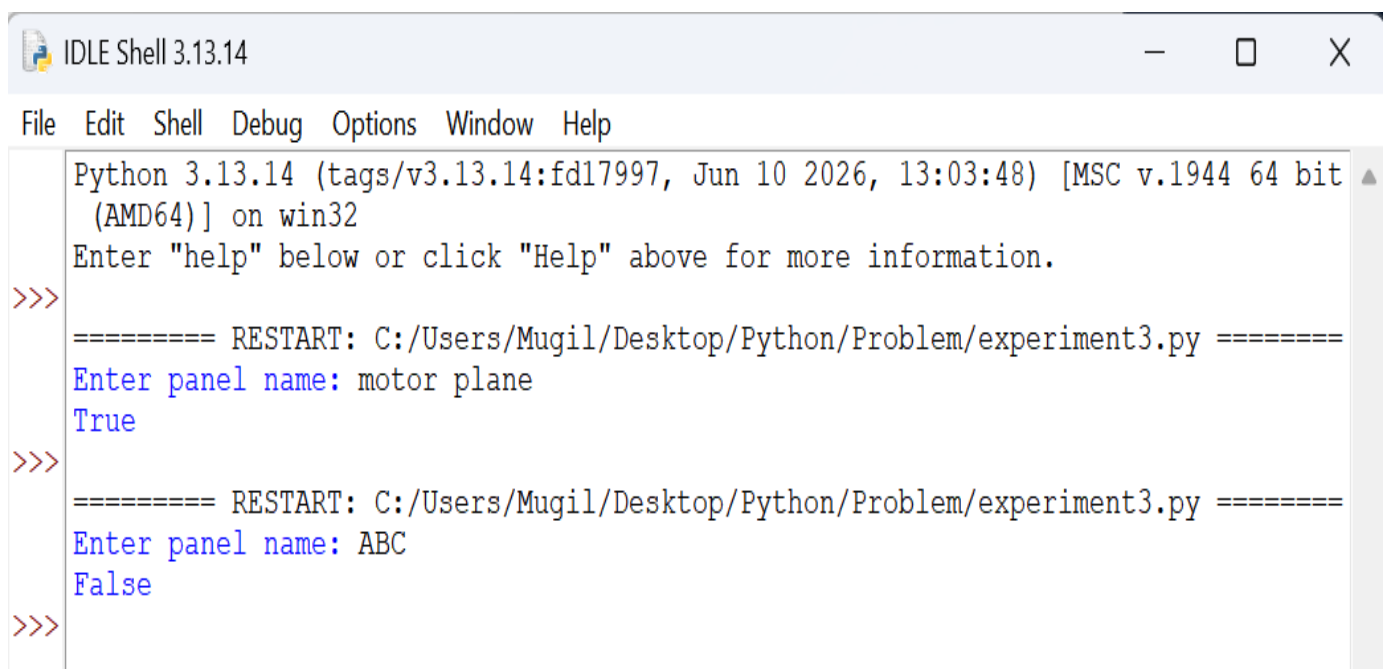
```

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Mugil/Desktop/Pyt
hon/Experiment/Experiment3.py =====
Enter first word: motor
Enter second word: panal
False
>>>
===== RESTART: C:/Users/Mugil/Desktop/Pyt
hon/Experiment/Experiment3.py =====
Enter first word: apple
Enter second word: plane
True
>>>

```

APPLICATION PROBLEM CODING

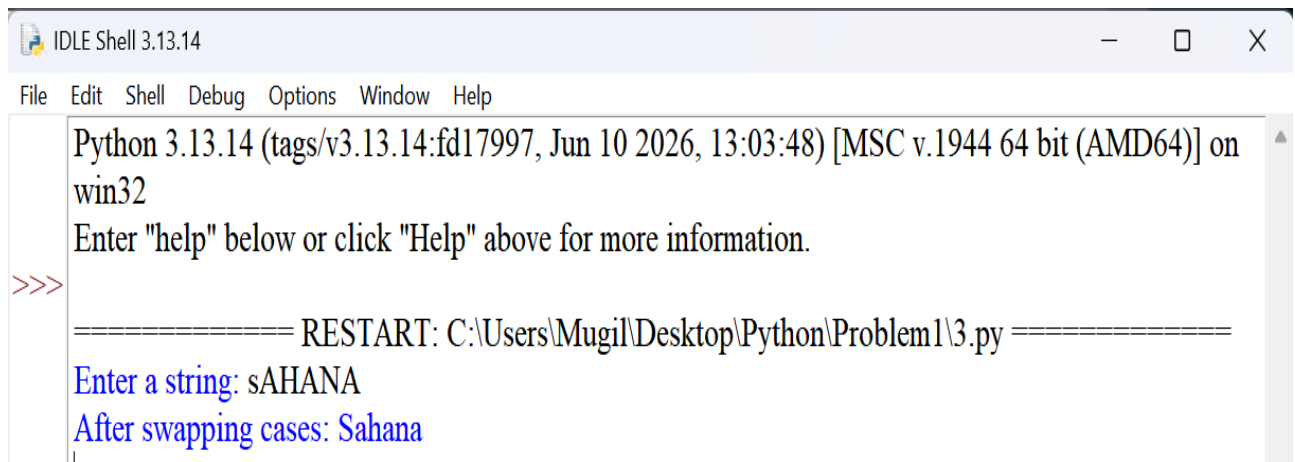
```
def duplicate_letters(word):  
    for i in word:  
        if word.count(i) > 1:  
            return True  
    return False  
panel = input("Enter panel name: ")  
print(duplicate_letters(panel))
```

OUTCOMES OF APPLICATION PROGRAM CODING

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit  
(AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
>>> ===== RESTART: C:/Users/Mugil/Desktop/Python/Problem/experiment3.py =====  
Enter panel name: motor plane  
True  
>>> ===== RESTART: C:/Users/Mugil/Desktop/Python/Problem/experiment3.py =====  
Enter panel name: ABC  
False  
>>>
```

PROBLEM CODING

```
text = input("Enter a string: ")
result = text.swapcase()
print("After swapping cases:", result)
```

OUTCOMES OF PROGRAM CODING

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on
win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem1\3.py =====
Enter a string: sAHANA
After swapping cases: Sahana
```

VIVA QUESTION**1. What is the difference between / and // in Python?**

- / is used for floating-point division and returns a decimal value. Example: $5 / 2 = 2.5$
- // is used for integer (floor) division, returning only the integer part by discarding the decimal part. Example: $5 // 2 = 2$

2. How is Exceptional handling done in Python?

Exception handling is done using the try, except, finally blocks.

try:

```
x = 1 / 0
```

except ZeroDivisionError:

```
print("Cannot divide by zero!")
```

finally:

```
print("Execution finished.")
```

3. What is swapcase function in Python?

The swapcase() function returns a string with uppercase letters converted to lowercase and vice versa

"Hello World".swapcase() # Output: "hELLO wORLD"

4. Difference between for loop and while loop in Python.

- for loop: Iterates over a sequence (like a list or string) and runs a set of statements for each item in the sequence.
- while loop: Repeats a block of code as long as a condition is true.

5. Can we pass a function as an argument in Python?

Yes, functions can be passed as arguments to other functions in Python.

RESULT:

Thus, the duplicate letter in a word of the given sentence is done using python programming and it is applied to identify and display true or false. Also, its application program was also executed and verified successfully.

INFERENCE:

The experiment demonstrates that Python functions and string operations can effectively detect duplicate letters in words within a sentence. The program accurately returns True or False based on the presence of duplicate letters, validating the implemented logic and its usefulness in text processing and data validation applications.

Assessment	Max. Marks	Marks Obtained
Conduction of Practical	15	
Report	5	
Viva	5	
Total Marks	25	
Signature of Lab In-Charge		